Christmastime is Christmas Math Time!

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1 The Infinite Gift and Gabriel's Horn

Algebraic geometry is populated by all kinds of crazy objects. Being that it is Christmastime (and so Christmas Math Time!) please check out the paradoxical object shown in Figure 1. This object is sometimes known as The Infinite Gift.

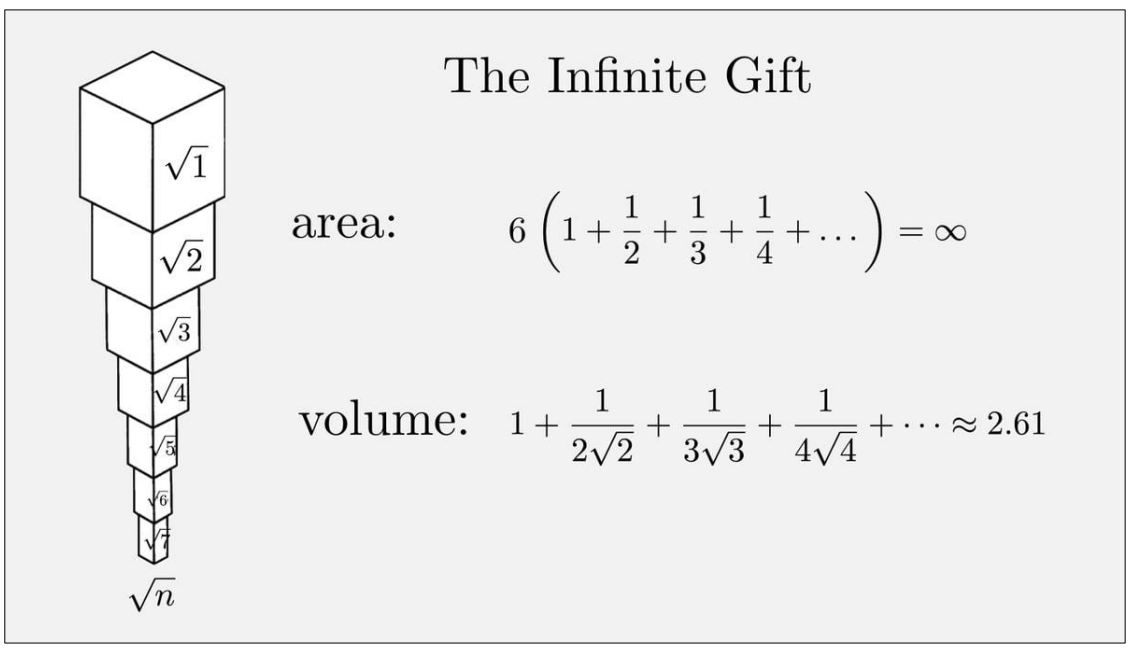


Figure 1: The Infinite Gift

In the Infinite Gift the length of the side of the nth box is $\frac{1}{\sqrt{n}}$ so the area of a side of the nth box is $\left(\frac{1}{\sqrt{n}}\right)^2 = \frac{1}{n}$. Since a box has 6 sides the surface area of the nth box is $6 * \frac{1}{n}$. Then what you find is that in the limit (as $n \to \infty$) the Infinite Gift has infinite surface area but finite volume!

Interesting aside: In the limit the area of the Infinite Gift equals 6 times the harmonic series (which we know diverges) [6].

The continuous version of this object is known as Gabriel's Horn (also known as Torricelli's Trumpet), which is shown in Figure 2.

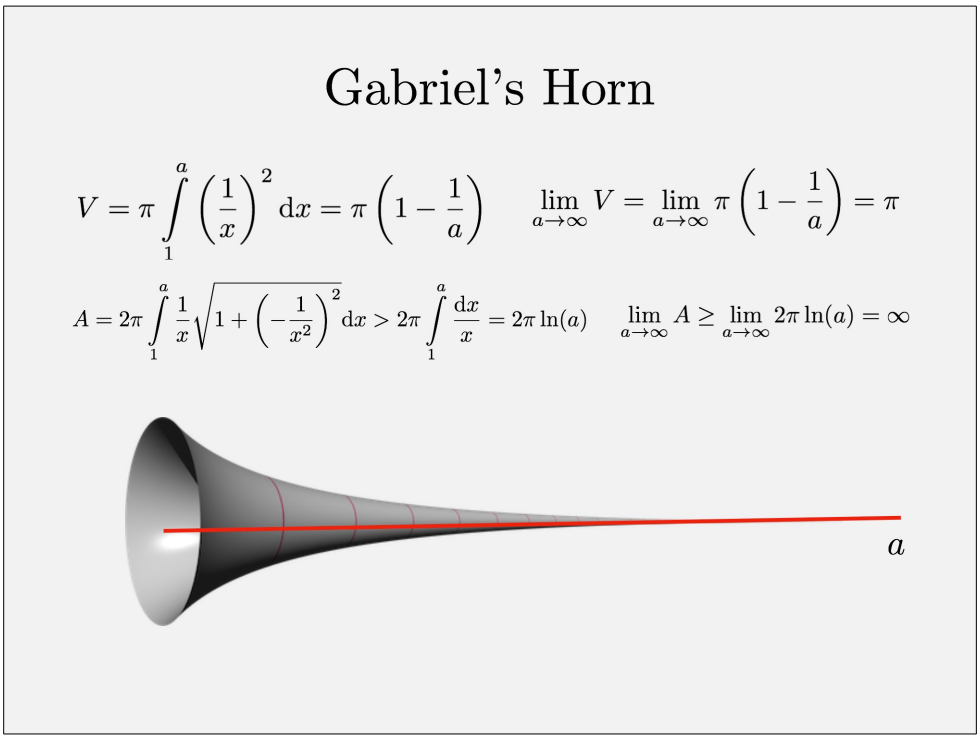


Figure 2: Gabriel's Horn

As we can see in Figure 2, Gabriel's Horn has volume π and area ∞ in the limit. These properties lead to an interesting situation known as the Painter's Paradox [1, 2].

1.1 The Painter's Paradox

This is the Painter's Paradox: Somehow even though you can fill Gabriel's Horn with paint (its volume is finite) you still won't have enough paint to paint its inside surface (its area is infinite)!

2 Fermat's Christmas Theorem

Fermat's Christmas Theorem (aka Fermat's theorem on sums of two squares) [3] is a beautiful and simple theorem which states that an odd prime number p can be expressed as

$$p = r^2 + s^2$$

where $r, s \in \mathbb{N}$, if and only if $p \equiv 1 \pmod 4$. That is, the theorem holds iff $p = 4n + 1$ for some $n \in \mathbb{N}$.

For example, the primes 5, 13, 17, 29, 37 and 41 are all congruent to 1 modulo 4 and can be expressed as sums of two squares in the following ways:

$$
\begin{aligned}
5 &= 1^2 + 2^2 \\
13 &= 2^2 + 3^2 \\
17 &= 1^2 + 4^2 \\
29 &= 2^2 + 5^2 \\
37 &= 1^2 + 6^2 \\
41 &= 4^2 + 5^2
\end{aligned}
$$

On the other hand, the primes 3, 7, 11, 19, 23 and 31 are all congruent to 3 modulo 4 and none of them can be expressed as the sum of two squares. This is the easier part of the theorem since it follows immediately from the observation that all squares are congruent to 0 or 1 modulo 4.

The prime numbers p for which Fermat's Christmas Theorem is true are called Pythagorean primes. See [5] for more on Pythagorean primes.

BTW, this theorem is called Fermat's Christmas Theorem because Fermat announced a proof of the theorem in a letter to Mersenne dated December 25, 1640. And of course, Fermat didn't include a proof in the letter.

A variety of proofs of Fermat's Christmas Theorem can be found in [4].

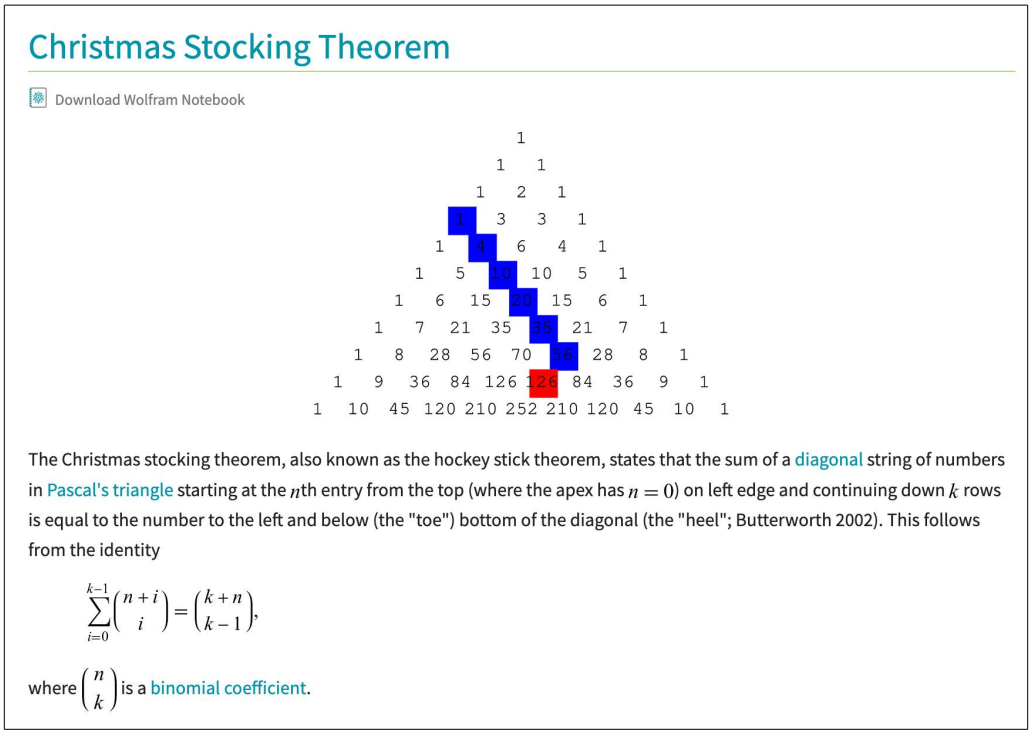
3 The Christmas Stocking Theorem

"The Christmas stocking theorem, also known as the hockey stick theorem, states that the sum of a diagonal string of numbers in Pascal's triangle starting at the nth entry from the top (where the apex has n=0) on left edge and continuing down k rows is equal to the number to the left and below (the "toe") bottom of the diagonal (the "heel"; Butterworth 2002)." [7].

That is, Christmas Stocking Theorem states that for all $n \in \mathbb{N}$ we have

$$\sum_{i=0}^{k-1} \binom{n+i}{i} = \binom{k+n}{k-1}$$

where $\binom{n}{k}$ is a binomial coefficient (aka "n choose k"). This (and the Christmas stocking) is shown in Figure 3.



Christmas Stocking Theorem

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```
                1
              1   1
            1   2   1
          1   3   3   1
        1   4   6   4   1
      1   5  10  10   5   1
    1   6  15  20  15   6   1
  1   7  21  35  35  21   7   1
1   8  28  56  70  56  28   8   1
1   9  36  84 126 126  84  36   9   1
1  10  45 120 210 252 210 120  45  10   1
```

The Christmas stocking theorem, also known as the hockey stick theorem, states that the sum of a diagonal string of numbers in Pascal's triangle starting at the nth entry from the top (where the apex has $n = 0$) on left edge and continuing down k rows is equal to the number to the left and below (the "toe") bottom of the diagonal (the "heel"; Butterworth 2002). This follows from the identity

$$\sum_{i=0}^{k-1} \binom{n+i}{i} = \binom{k+n}{k-1},$$

where $\binom{n}{k}$ is a binomial coefficient.

Figure 3: The Christmas Stocking Theorem

4 Merry X-mas!

$$
\begin{aligned}
y &:= \frac{\ln\left(\frac{x}{m} - sa\right)}{r^2} && \text{\# define } y \\
&\Rightarrow r^2 y = \ln\left(\frac{x}{m} - sa\right) && \text{\# multiply both sides on the left by } r^2 \\
&\Rightarrow e^{r^2 y} = e^{\ln\left(\frac{x}{m} - sa\right)} && \text{\# } x = y \Rightarrow b^x = b^y \text{ ; exponentiate with } b = e \\
&\Rightarrow e^{r^2 y} = \frac{x}{m} - sa && \text{\# } e^{\ln(x)} = x \\
&\Rightarrow e^{r^2 y} = \frac{x}{m} - as && \text{\# assume multiplication is commutative } (sa = as) \\
&\Rightarrow me^{r^2 y} = x - mas && \text{\# multiply both sides on the left by } m \\
&\Rightarrow \mathbf{me^{rry} = x - mas} && \text{\# } r^2 = rr \Rightarrow \textbf{Merry X-mas!}
\end{aligned}
$$

LaTeX Source

`https://www.overleaf.com/read/fwsfybtfmryr`

References

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[3] Wikipedia Contributors. Fermat's theorem on sums of two squares — Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Fermat%27s_theorem_on_sums_of_two_squares&oldid=990575323, 2020. [Online; accessed 26-December-2020].

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[7] Wolfram MathWorld. Christmas Stocking Theorem. https://mathworld.wolfram.com/ChristmasStockingTheorem.html, 2017. [Online; accessed 25-December-2022].